

Eidi Digital Bond Program (EDBP)

White Paper: Trend Analysis, Policy Framework & Implementation Roadmap

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Capacity:

Prepared for: State Bank of Pakistan, Ministry of Finance, Commercial Banks, Fintechs, Academia & Development Partners
(as prospective stakeholders - not as affiliated institutions)

> Independence notice: This white paper is the independent work of the author named above. It i

s not prod

Document Control

| Field | Detail |

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1. The Problem We Are Solving

Before describing any solution, this paper establishes the core problem: Pakistan's Eid season

triggers a

1.1 What Happens Every Eid - A Predictable Cash Shock

Each year, in the weeks before Eid-ul-Fitr and Eid-ul-Adha, millions of Pakistani households follow the same pattern:

1. Withdraw large sums of cash from banks and ATMs

2. Obtain

This is not a marginal behaviour. Historical reporting places the scale at:

| Indicator | Reported Scale | Year / Source |

|-----|

Note: SBP does not publish a real-time official Eid withdrawal series. Figures reflect historical press and research estimates and should be re-baselined during a pilot.

The result is ATM congestion, branch overcrowding, accelerated banknote printing, informal premium trading of fresh notes,

and seasonal liquidity pressure on banks - documented in academic research on Ramadan/Eid deposit volatility in Pakistan (Limodio, UZH).

1.2 The Behavioural Problem - Cash Leaves the Formal System

The problem is not only logistical. It is behavioural and structural:

Step 1 - Deliberate withdrawal from banks. Around Eid, many account holders withdraw cash *spec

ificantly* to

Step 2 - Exit from the formal financial system. Once withdrawn, Eidi cash sits outside bank accounts, EMI wallets, and savings products. It is held in homes, pockets, and envelopes - unrecorded, uninsured, and invisible to the financial system.

Step 3 - Consumption, not retention. Eidi cash is overwhelmingly spent, not saved. Children and

recipients

Step 4 - A missed financial-inclusion moment. Eid is the one occasion each year when virtually

every Pak

BANK ACCOUNT

CASH IN HAND

RETAIL / INFORMAL

???????

This is the problem EDBP targets: not the tradition of Eidi, but the cash pipeline that strips value from the formal system at the exact moment when digital onboarding would be most culturally acceptable.

1.3 Systemic Costs of the Status Quo

Cost Category	Impact
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1.4 Why Current Approaches Do Not Solve This

Approach	Limitation
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No existing instrument combines: cultural Eidi framing + government guarantee + instant digital delivery + child recipient onboarding + diaspora purchase capability + optional savings conversion.

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2. Executive Summary

Having established the problem above, this paper proposes the Eidi Digital Bond Program (EDBP)

- a cultura

This is not monetary expansion. Citizens purchase bonds using existing digital money. The State Bank receives upfront liquidity. Redemption releases funds already in the system. No new currency is printed.

Why Now - Five Converging Trends (2024-2026)

| Trend | Signal | Relevance to EDBP |

|-----|

Estimated Impact (Conservative)

| Metric | Estimate | Basis |

|-----|

Recommendation

Launch EDBP as a regulated pilot under SBP supervision in partnership with CDNS, integrated with RAAST and distributed through licensed banks and EMIs - timed to Ramadan/Eid 2027. Scale nationally upon KPI validation.

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1. The Problem We Are Solving

2. Executi

3. Introduction & Cultural-Economic Context

3.1 What Is Eidi?

Eidi (????) is a cherished Islamic cultural practice of giving cash gifts to children, younger

relatives, (

This cultural universality makes Eidi a uniquely powerful financial inclusion lever: every household participates; every child is a potential recipient; every adult is a potential sender.

3.2 The Missed Opportunity

Pakistan has invested heavily since 2019 in digital payments infrastructure - RAAST, Asaan Digi

tal Account

- Citizens queue at bank branches for fresh notes

- ATMs are

EDBP reframes Eidi from a cash logistics problem into a national financial capability moment.

3.3 Program Thesis

> Replace the medium of Eidi - not the tradition of Eidi.

An Eidi Digital Bond preserves the ritual of giving while routing value through regulated digital rails, creating traceability, safety, optional savings, and temporary sovereign financing - without expanding money supply.

4. Problem Deep-Dive: Data & Behavioural Economics

This section supplements §1 with additional data and behavioural analysis. The core problem - cash extraction, informal circulation, and consumption rather than savings - is established upfront in Section 1.

4.1 Seasonal Cash Demand - Historical Data

Pakistan's Eid economy generates substantial, predictable cash extraction from the banking system:

| Indicator | Reported Scale | Source Context |

|-----

Note: SBP does not publish real-time Eid-specific withdrawal series; figures reflect historical press and research estimates. A pilot should establish baseline telemetry.

4.2 Operational Strain on Infrastructure

The Eid cash shock creates systemic friction:

1. ATM congestion - Long queues, out-of-cash events, urban branch overcrowding

2. Bank li
cycles)

4.3 Behavioural Economics of Eid Cash

Research and market observation confirm a consistent behavioural loop:

| Stage | Behaviour | Consequence |

|-----|

Academic evidence from Pakistan shows that Ramadan and Eid cause measurable deposit volatility

at banks, v

EDBP interrupts this loop by keeping Eidi value inside regulated digital wallets until the recipient chooses to cash out or save - converting a consumption event into a potential inclusion and savings event.

4.4 Currency Management Cost Context

Pakistan's currency management costs are material and rising:

- PKR 10 note printing alone: PKR 8-10 billion annually (~35% of all notes printed)

- 10-year

EDBP does not replace denomination reform - it complements it by reducing seasonal demand for physical notes across all denominations used in Eidi (PKR 50-5,000).

4.5 Financial Inclusion Gap

Despite progress (16% -> 64% bank account ownership, 2015-2023), Pakistan's NFIS 2024-28 acknowledges a persistent barrier:

> *"General preference for cash embedded in the culture."*

> - NFIS 2

Eidi is the exception where digital giving can be socially encouraged without fighting cultural cash preference - because the sender wants convenience and the receiver wants instant value.

5. Global & Regional Trend Analysis

5.1 Trend 1 - Tokenized Government Bonds for Retail Investors

Status: Accelerating globally; emerging markets leapfrogging legacy bond market infrastructure.

| Jurisdiction | Initiative | Key Feature | Lesson for Pakistan |

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The Philippines model is the closest analog: 94 million GCash users can access government bonds from PHP 500 through an app they already use. Pakistan's JazzCash (40M+ users) and Easypaisa (35M+ active) provide equivalent distribution scale.

Strategic insight: EDBP is not experimental in a global vacuum - it is a localized, seasonal, micro-denomination variant of a validated policy trend.

5.2 Trend 2 - Digital Vouchers & Purpose-Bound Money

| Jurisdiction | Program | Mechanism |

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Common thread: Central banks distinguish between CBDC (money creation) and digital vouchers/bonds (existing money, programmed use). EDBP falls in the latter category - a critical distinction for SBP monetary policy credibility.

5.3 Trend 3 - Instant Payment Rails as National Infrastructure

Pakistan's RAAST places the country among leading instant-payment adopters:

| Metric | RAAST Performance |

|-----|

RAAST's Bulk Payments module - already identified by CDNS for profit payments and investment encashment - is the natural settlement backbone for EDBP redemption.

5.4 Trend 4 - Mobile Wallets as Primary Bank for the Masses

| Platform | Scale | Strategic Role in EDBP |

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Trend conclusion: Pakistan has achieved distribution scale before achieving cultural cash displacement. EDBP targets the cultural moment.

5.5 Trend 5 - Scrippless Government Savings (Domestic Precedent)

Pakistan is already transitioning government savings instruments to digital:

- Digital Prize Bonds (Registered) Rules, 2024 - Notified by Finance Division

- CDNS m

EDBP should be architected as a sibling instrument to Digital Prize Bonds - not a competing silo - sharing KYC, registry, and settlement infrastructure.

6. Pakistan Macro Trends & Policy Alignment

6.1 NFIS 2024-28 Alignment Matrix

| NFIS Goal | EDBP Contribution |

|-----|

6.2 SBP "Go Cashless" & Financial Literacy Week

SBP Deputy Governor Saleem Ullah's 2025 "Go Cashless" drive explicitly targets behavioral shift from cash to digital. EDBP provides a concrete product for that campaign - not just awareness messaging.

6.3 Currency Reform Complementarity

The 2026 SBP/PSPC recommendation to replace PKR 10 notes with coins (saving PKR 40-50B over 10 years) addresses structural note cost. EDBP addresses seasonal note demand. Together, they form a coherent currency modernization agenda.

6.4 Political Economy Considerations

EDBP is politically viable because it:

- Respects Islamic cultural practice (not a "war on cash")

- Benefits

7. Who Benefits - Stakeholder Value Propositions

EDBP creates value across the entire financial ecosystem. This section details how each stakeholder benefits - not just that they benefit.

7.1 Children & Eidi Recipients

| Benefit | Detail |

|-----

7.2 Families (Senders & Households)

| Benefit | Detail |

|-----

7.3 Overseas Pakistanis (Diaspora)

| Benefit | Detail |

|-----

7.4 Government of Pakistan

| Benefit | Detail |

|-----

7.5 State Bank of Pakistan & Financial System

| Benefit | Detail |

|-----

7.6 Banks, EMIs & Fintechs

| Benefit | Detail |

|-----

7.7 Broader Economy

| Benefit | Detail |

|-----

8. The Eidi Digital Bond - Instrument Design

8.1 Definition

An Eidi Digital Bond (EDB) is a short-duration, scripless, government-guaranteed digital savings instrument issued specifically for Eid gifting, denominated in Pakistani Rupees, transferable between KYC-verified wallets, and redeemable at par during designated Eid windows via RAAST.

8.2 Core Properties

| Property | Specification |

|-----

8.3 What EDB Is - and Is Not

| EDB Is | EDB Is Not |

|-----

8.4 Comparison to Adjacent Instruments

| Feature | EDB | Digital Prize Bond | RAAST P2P Transfer | Prepaid Gift Card |

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9. Program Architecture & User Journey

9.1 Three-Step Flow

???????????????? ????????????????? ?????????????????

? 1. BUY

9.2 Detailed User Journey - "Ali & Sara"

| Step | Actor | Action | Backend Event |

|-----|

Total time: < 60 seconds per transaction. No ATM. No branch. No cash handling.

9.3 Distribution Channel Architecture

????????????????????

9.4 Child & Minor Wallet Framework

| Age Group | Account Type | Requirements |

|-----|

This structure supports NFIS youth inclusion goals and creates first account at first Eidi - a lifelong financial services entry point.

10. Overseas Pakistanis & Diaspora Eidi

A significant share of Eidi in Pakistan is funded by overseas Pakistanis - approximately 9-10 m

illion Pakis

EDBP extends the program to overseas Pakistanis, making diaspora Eidi a first-class use case rather than an afterthought.

10.1 Diaspora User Journey - "Ahmed in Dubai -> Niece Sara in Lahore"

| Step | Actor | Action | Backend Event |

|-----|-----|

No cash handoff. No hawala. No dependency on a relative visiting from abroad with physical notes.

10.2 Identity-Linked Assignment

Overseas senders assign bonds using the recipient's official identity:

| Recipient Type | Required Identifier | Guardian Rule |

|-----|-----|

NADRA e-KYC APIs verify identity at assignment time, preventing misdirected gifts and enabling AML compliance.

10.3 Funding Channels for Overseas Pakistanis

| Channel | Mechanism | Status |

|-----|-----|

10.4 Regulatory Considerations

- Overseas purchasers must complete KYC under SBP RDA/NRP rules (already required for account holders)

- EDB as transfers

10.5 Diaspora Impact Estimate

| Scenario | Assumption | Volume |

|-----|-----|

Diaspora Eidi alone could make EDBP one of the largest seasonal digital payment events in Pakistan's history.

10.6 Comparison - Diaspora Eidi Today vs. EDBP

| | Today (Cash/Hawala) | EDBP (Digital Bond) |

| -- |-----

11. Economic & Fiscal Analysis

11.1 Cost Avoidance Model

| Cost Category | Annual Estimate (PKR) | EDBP Displacement Assumption | Savings (PKR) |

|-----

Methodology note: Author’s original estimate of PKR 7-12B printing savings aligns with this model when Eid-related printing share is assumed at 25-40% of total note production costs.

11.2 Temporary Government Financing Benefit

Purchase-to-redemption float:

If 20 million citizens purchase an average of PKR 2,000 in EDBs during a 25-day Ramadan window, gross issuance = PKR 40 billion held by government until Eid redemption.

| Scenario | Issuance Volume | Avg. Hold Period | Implied Float |

|-----

At current SBP policy rate, even a zero-coupon instrument provides fiscal breathing room equivalent to a short-term T-bill auction - without auction friction.

Critical safeguard: Float must be ring-fenced in a segregated redemption reserve account at SBP, fully backed 1:1. This is not deficit financing.

11.3 Financial Inclusion Economic Returns

Research on financial inclusion consistently links account ownership to:

- Higher formal savings rates

- Improved

EDBP creates 2 touchpoints per gift (sender + receiver), doubling onboarding potential vs. a unilateral government campaign.

11.4 Fintech & Banking Revenue (Secondary Effects)

| Stakeholder | Revenue / Value Driver |

|-----

12. Monetary Policy & Inflation Safeguards

12.1 The Closed-Loop Principle

EDBP operates as a closed-loop monetary circuit:

1. Purchase: Digital money leaves citizen wallet -> enters government bond registry (SBP/CDNS account)

2. Gift: Bo

Net change in M0 (currency in circulation): Negative or neutral - never positive.

12.2 Distinction from Money Printing

SBP defines currency printing drivers as:

- Replacement of soiled notes

- Currency

EDBP involves none of these expansionary mechanisms. It is economically equivalent to:

> Citizens temporarily buying a 30-day T-bill and gifting the claim to a relative who redeems it on Eid.

12.3 SBP Safeguards (Recommended)

| Safeguard | Implementation |

|-----

13. Regulatory & Legal Framework

13.1 Proposed Legal Basis

| Instrument | Applicable Law / Rule |

|-----

13.2 Regulatory Architecture

Finance Division (Policy Owner)

?

13.3 EMI & Bank Participation Framework

Participating institutions should sign a Standard EDBP Participation Agreement covering:

- API integration specifications
- KYC reli

14. Technology & Interoperability

14.1 Core Technical Components

Component	Function	Existing Asset	

14.2 API-First Design (For Fintech Partners)

Minimum viable API endpoints:

- POST /edb/purchase - Buy bonds from wallet
- POST /edb

14.3 Security Standards

- End-to-end encryption for bond transfer messages
- Multi-fac

14.4 No Blockchain Required (Phase 1)

Unlike Philippines GBonds (DLT registry), Pakistan can launch EDBP on centralized CDNS registry integrated with RAAST - faster, cheaper, regulator-native. DLT may be evaluated in Phase 3 if interoperability with regional markets is desired.

15. Stakeholder Roles & Commercial Model

15.1 Stakeholder Matrix

Stakeholder	Role	Incentive	

15.2 Fee Structure (Recommended - Zero Consumer Cost)

| Party | Fee | Rationale |

|-----|

16. Risk Assessment & Mitigation

| # | Risk | Likelihood | Impact | Mitigation |

|---|-----|

17. Implementation Roadmap (18 Months)

Phase 0 - Policy Design (Months 1-3)

| Task | Owner | Deliverable |

|-----|

Phase 1 - Build & Integrate (Months 4-8)

| Task | Owner | Deliverable |

|-----|

Phase 2 - Pilot (Months 9-12)

| Parameter | Specification |

|-----|

Phase 3 - National Rollout (Months 13-18)

Phase 4 - Institutionalize (Month 18+)

- Annual NFIS reporting inclusion
- Academic reporting inclusion

18. KPIs, Monitoring & Research Agenda

18.1 Program KPIs

| KPI | Year 1 Target | Year 3 Target |

|-----|-----|

18.2 Research Questions for Academia

1. Does digital Eidi increase child savings rates vs. cash Eidi?
2. What is the impact of digital Eidi on child savings rates?

Suggested methodology: Difference-in-differences using pilot cities vs. control cities; partnership with SBP Research Department and LUMS/IBA/State Bank training institutes.

18.3 SBP Reporting Integration

EDBP metrics should be added to the Quarterly Payment Systems Review as a supplementary table - giving banking professionals and researchers standardized data series.

19. Optional Product Extensions

19.1 Eidi Savings Mode

- Recipient opts to hold EDB beyond Eid
- Auto-contribution to EDB

19.2 Donation Bonds

- Sender purchases EDB -> transfers to registered charity (Edhi, Shaukat Khanum, etc.)
- Charity receives EDB

19.3 Family Group Eidi

- Multiple family members contribute to a pooled EDB

- Recipien

19.4 Corporate Eidi Program

- Employers distribute EDBs to employees' children

- Bulk issu

20. Policy Recommendations

For Government (Ministry of Finance)

1. Approve EDBP in principle as a flagship NFIS 2024-28 initiative

2. Issue E

For State Bank of Pakistan

1. Charter an EDBP Task Force (Payments Systems Dept. + Banking Policy + Research)

2. Integrat

For Banks & Fintechs

1. API integration with CDNS EDB Gateway by Q4 2026

2. UX inve

For Researchers & Universities

1. Apply for SBP Research Department partnership for pilot evaluation

2. Publish

21. Conclusion

The Eidi Digital Bond Program sits at a rare intersection: cultural tradition, fiscal discipline, financial inclusion, and existing digital infrastructure - all aligned in the same direction.

Pakistan has already built the rails (RAAST), the wallets (80M+ accounts), the regulatory frame

work (Digi

EDBP is that product.

For banking professionals, it represents a scalable, low-risk, high-volume seasonal product wit

h clear int

The infrastructure is built. The culture is ready. The diaspora is connected. The economics are favorable. The time to pilot is now.

22. Data Sources & Methodology

This section documents every category of statistic used in this paper, how it was obtained, and whether it is published data or a model estimate.

22.1 Data Classification

Label	Meaning	
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22.2 Published Statistics - Key Figures & Sources

RAAST & Digital Payments

Stat	Value	Year	Source	
------	-------	------	--------	--

Financial Inclusion

Stat	Value	Year	Source	
------	-------	------	--------	--

Eid Economy & Seasonal Cash

Stat	Value	Year	Source	
------	-------	------	--------	--

Currency & Printing

Stat	Value	Year	Source	
------	-------	------	--------	--

Fintech & Wallets

| Stat | Value | Year | Source |

|-----|-----|

Diaspora & Remittances

| Stat | Value | Year | Source |

|-----|-----|

Global Benchmarks

| Stat | Value | Year | Source |

|-----|-----|

22.3 Model Estimates (No Single Official Source)

| Stat | Value | Methodology |

|-----|-----|

22.4 Primary Official Portals for Updated Data

| Institution | Link |

|-----|

22.5 Limitations & Caveats

1. No official Eid withdrawal time series - SBP does not publish real-time Eid-specific data; historical press estimates (2017) are the best public benchmarks.

2. Model e

23. About the Author

Muhammad Ammar Jamshed

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Muhammad Ammar Jamshed is the sole author of this white paper, written entirely in a personal and independent capacity. He is not employed by, contracted to, or representing the State Bank of Pakistan, the Government of Pakistan, any bank, fintech,

or other institution referenced in this document.

Background

Muhammad Ammar Jamshed holds an MSc in Data Science from the University of London. His professional work spans data analytics, financial systems thinking, and public-policy research - with a focus on applying quantitative methods to real-world economic and inclusion challenges in emerging markets.

He has worked across data science, analytics, and entrepreneurship - including experience in consumer goods analytics and independent consulting - and writes on data science and policy topics for a general professional audience.

Why This Paper

The Eidi Digital Bond Program emerged from observing a recurring gap in Pakistan's otherwise ac

celerating

Author's Role in This Work

| Aspect | Detail |

|-----|

Independence Statement

> This white paper is an independent scholarly and policy contribution. References to the State

Bank of P

Relevant Expertise Applied

- Data analysis & modelling - impact estimates, cost-avoidance scenarios, KPI frameworks

- Payment

Annexes

Annex A - Glossary

| Term | Definition |

|-----|

Annex B - Denomination Ladder Rationale

Annex C - Pilot City Selection Criteria

1. High mobile wallet penetration
2. RAAST

Annex D - Sample Regulatory Text (Draft Clause)

> *"The Federal Government may issue Eidi Digital Bonds in scripless, registered form for the p

urpose of

Annex E - Implementation Checklist (Quick Reference)

- [] Cabinet approval in principle
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Academic

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